

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Digital Radio Lab

Transmission and Reception of Standard Waveforms (Sine, Cosine, Triangular, Sawtooth, Square) using GNU Radio & PlutoSDR

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Experiment	LAB Record - 1 (Part 2)

1. Aim

To transmit and receive five standard test waveforms — Sine, Cosine, Triangular, Sawtooth and Square — using GNU Radio Companion (GRC) and the ADALM-PLUTO Software Defined Radio (SDR), and to observe and compare the transmitted and received waveforms in both the time domain and the frequency domain.

2. Software Used

- GNU Radio Companion (GRC) 3.10 – QT GUI
- Python 3 (auto-generated GRC flowgraph back-end)
- ADALM-PLUTO SDR (PlutoSDR) – used as both transmitter (Sink) and receiver (Source)
- Ubuntu Linux (host OS for GNU Radio)
- libiio / libad9361-iio drivers for PlutoSDR IIO context

3. Theory

A Signal Source block in GNU Radio can generate several standard periodic waveforms which are widely used as test signals for verifying transmit/receive chains:

- Sine and Cosine waves: single-frequency (single-tone) signals whose spectrum ideally consists of one impulse at the signal frequency; they are the purest test signals for checking the fidelity of an SDR link.
- Triangular wave: contains only odd harmonics of the fundamental, with harmonic amplitude falling off as $1/n^2$, so its spectrum shows fast-decaying side-tones around the fundamental.
- Sawtooth wave: contains both even and odd harmonics with amplitude falling off as $1/n$, producing a dense 'comb' of harmonics that decays slowly — the richest harmonic content among the five waveforms.
- Square wave: contains only odd harmonics with amplitude falling off as $1/n$, giving a comb of harmonics that decays more slowly than the triangular wave but is sparser than the sawtooth wave (only odd multiples of the fundamental).

In this experiment, each waveform generated by the Signal Source is passed through a Throttle block (to pace the sample stream to the set sample rate) and then transmitted over the air using the PlutoSDR acting as a transmitter (Sink). The same waveform is simultaneously received back using a second/looped-back PlutoSDR acting as a receiver (Source). QT GUI Time Sinks display the transmitted and received waveforms in the time domain, while QT GUI Frequency Sinks display their spectra, allowing a direct comparison between the ideal (transmitted) and the SDR-received version of each waveform.

4. Source Code

This experiment was implemented entirely in GNU Radio Companion (GRC) using its graphical, drag-and-drop block-diagram environment — the Signal Source, Throttle, PlutoSDR Sink/Source, and QT GUI Time/Frequency Sink blocks were placed and wired together directly on the GRC canvas (Section 6, Fig. 1). No manual Python/C++ coding was performed by the student. GRC internally auto-generates the equivalent Python flowgraph from this block diagram purely for execution on the back-end; since the implementation itself is block-diagram based, the auto-generated code is not reproduced here. The same single flowgraph

was reused for all five waveforms by simply changing the Waveform parameter of the Signal Source block (Cosine → Sine → Triangle → Saw Tooth → Square); no other block or connection was modified.

5. Procedure

1. Open GNU Radio Companion and create a new flowgraph; set Generate Options to QT GUI.
2. Add global variable blocks: samp_rate = 1 M, center_freq = 1 G, signal_freq = 10 k and amplitude = 1.
3. Place a Signal Source block with Sample Rate = 1 M, Frequency = 1 kHz, Amplitude = 1, Offset = 0 and Initial Phase = 0, and set its Waveform parameter to Cosine.
4. Connect the Signal Source to a Throttle block (Sample Rate = 1 M, Limit = None) to pace the generated samples to the real-time sample rate.
5. From the Throttle block output, branch the signal three ways into: a QT GUI Frequency Sink (name 'Cosine', FFT Size 1024, Bandwidth 1 M) to view the transmitted spectrum, a QT GUI Time Sink (name 'cosine', 1.024k points) to view the transmitted waveform, and a PlutoSDR Sink (IIO 192.168.2.1, LO Frequency 2.4 GHz, Sample Rate 1 M, Buffer size 32.768k, Attenuation TX1 = 10 dB) to transmit the waveform over the air.
6. Configure a PlutoSDR Source block (IIO 192.168.2.1, LO Frequency 2.4 GHz, Sample Rate 1 M, Gain Mode Slow Attack, RF Bandwidth 20 MHz) to receive the transmitted signal.
7. Connect the PlutoSDR Source output to a QT GUI Time Sink (name 'Cosine received') and a QT GUI Frequency Sink (name 'cosine received') to display the received waveform and spectrum.
8. Run the flowgraph and record the transmitted and received time-domain and frequency-domain plots for the Cosine waveform.
9. Without changing any connection, open the Signal Source block properties and change the Waveform parameter to Sine; re-run the flowgraph and record the plots.
10. Repeat the previous step for Waveform = Triangle, Waveform = Saw Tooth and Waveform = Square, recording the corresponding transmitted/received time- and frequency-domain plots for each.
11. Compare the time-domain shape and the harmonic content visible in the frequency-domain plots across all five waveforms.

6. Observations

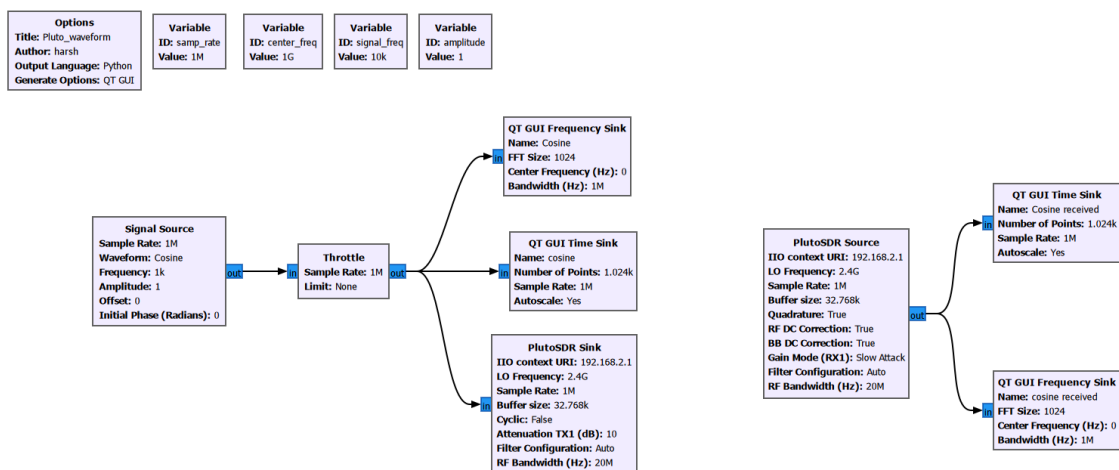


Fig. 1: GRC block diagram used for all five waveforms — Signal Source → Throttle → PlutoSDR Sink (Tx) / QT GUI Tx sinks, and PlutoSDR Source (Rx) → QT GUI Rx sinks. Only the Waveform parameter of the Signal Source block is changed between runs.

6.1 Cosine Wave

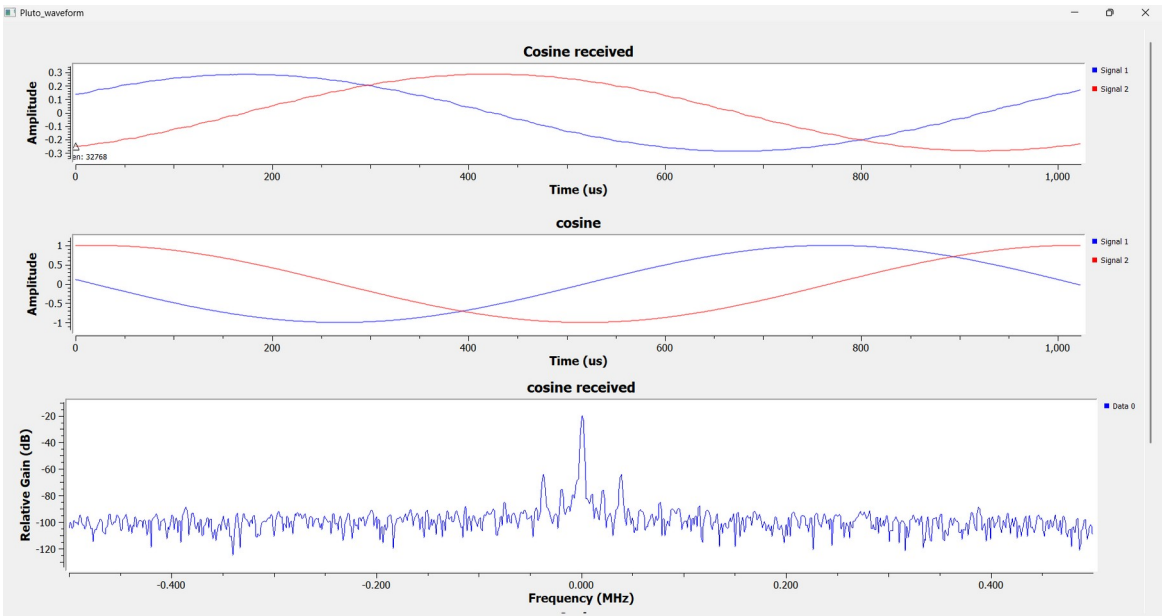


Fig. 2: Cosine wave — (top) received waveform 'Cosine received', (middle) transmitted waveform 'cosine', (bottom) received spectrum 'cosine received' showing a single dominant tone near 0 Hz.

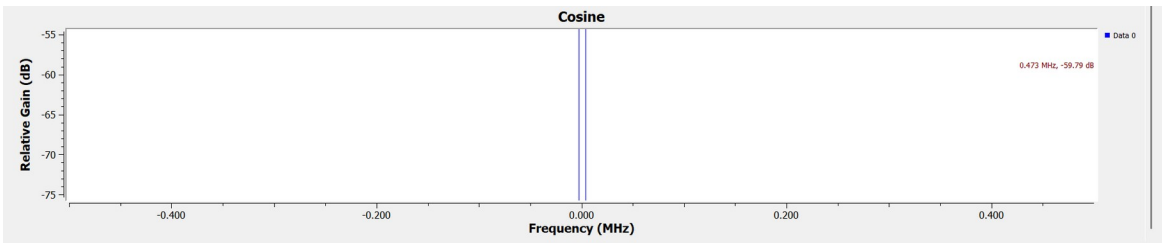


Fig. 3: Transmitted spectrum of the cosine wave (zoomed). Apart from the main tone, the spectrum is flat, with a marker at 0.473 MHz reading -59.79 dB, confirming a near-ideal single-tone spectrum.

6.2 Sine Wave

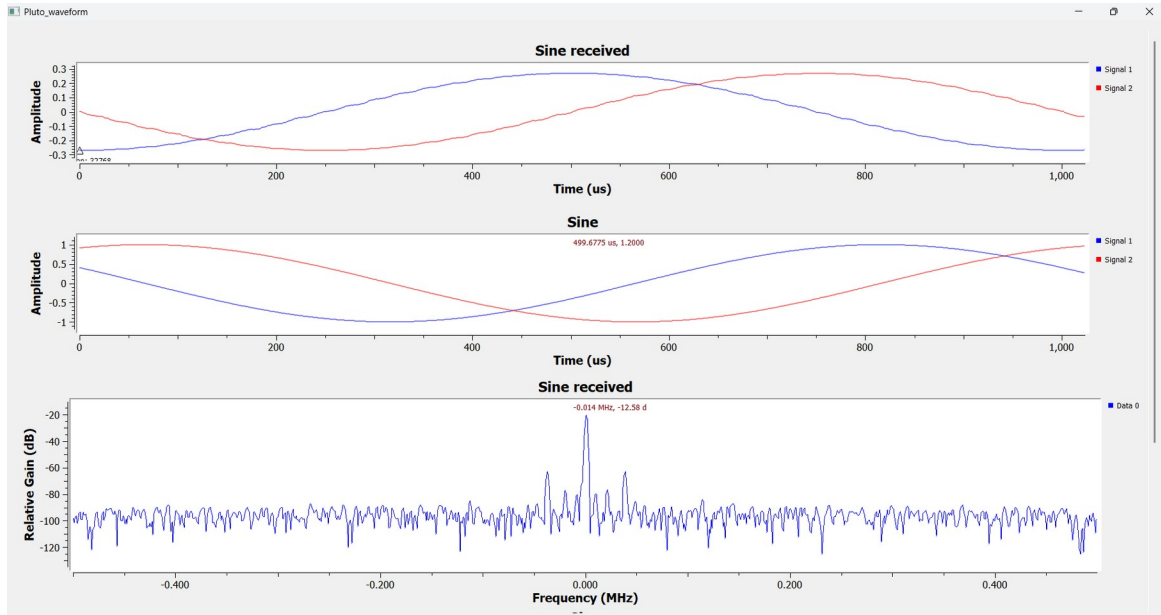


Fig. 4: Sine wave — (top) received waveform, (middle) transmitted waveform, (bottom) received spectrum with marker at -0.014 MHz, -12.58 dB showing the dominant tone close to 0 Hz.

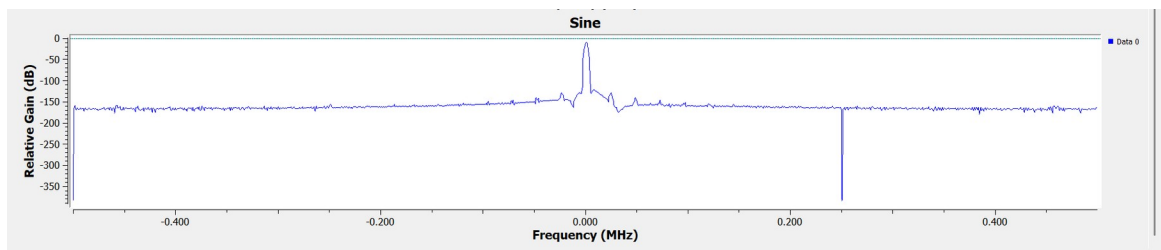


Fig. 5: Transmitted spectrum of the sine wave (zoomed), showing a sharp main tone near 0 Hz and a very deep null (down to about -350 dB) around 0.25 MHz.

6.3 Triangular Wave

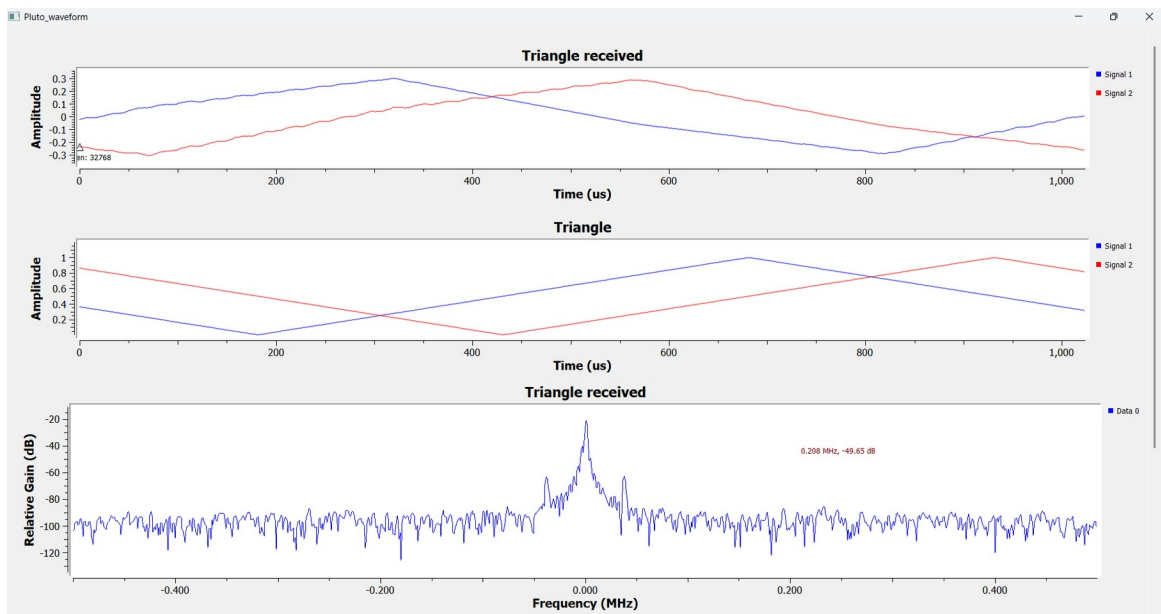


Fig. 6: Triangular wave — (top) received waveform showing the characteristic linear rise/fall shape, (middle) transmitted waveform, (bottom) received spectrum with marker at 0.208 MHz, -49.65 dB showing decaying harmonic side-tones.

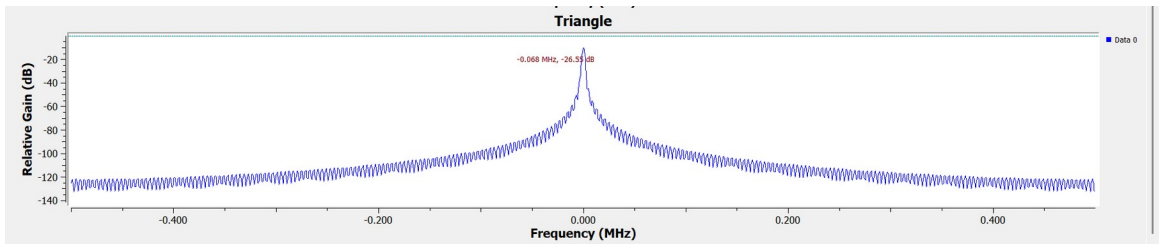


Fig. 7: Transmitted spectrum of the triangular wave (zoomed). Marker at -0.068 MHz, -26.5 dB. A series of harmonic side-lobes decaying away from the main tone is clearly visible, consistent with the $1/n^2$ harmonic roll-off of a triangular wave.

6.4 Sawtooth Wave

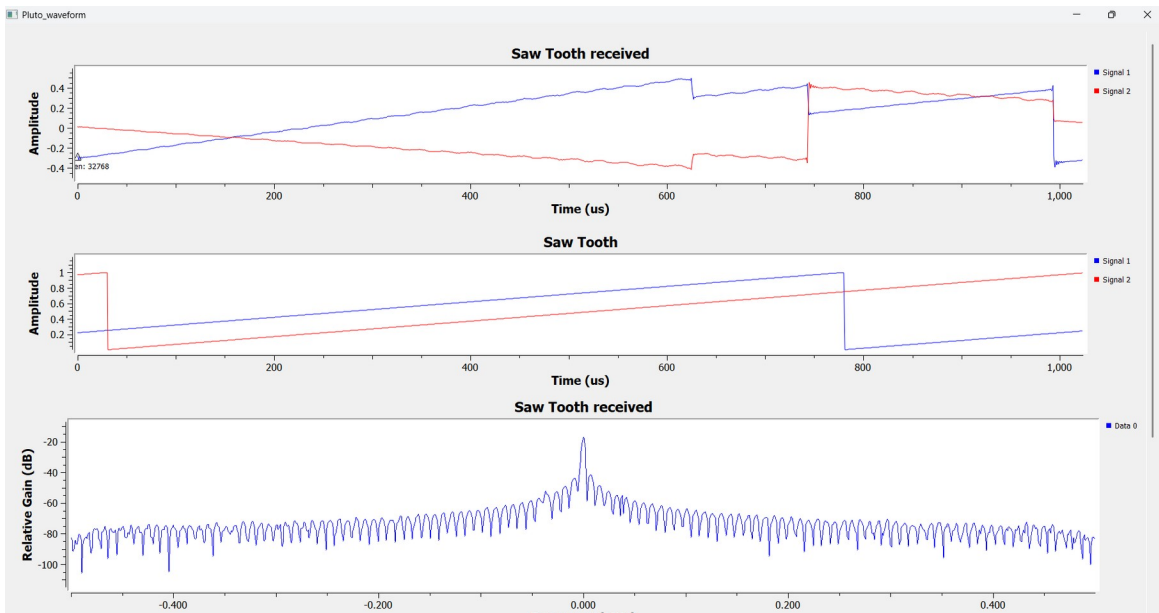


Fig. 8: Sawtooth wave — (top) received waveform showing the characteristic linear ramp with sharp fly-back, (middle) transmitted waveform, (bottom) received spectrum showing a dense comb of many harmonics.

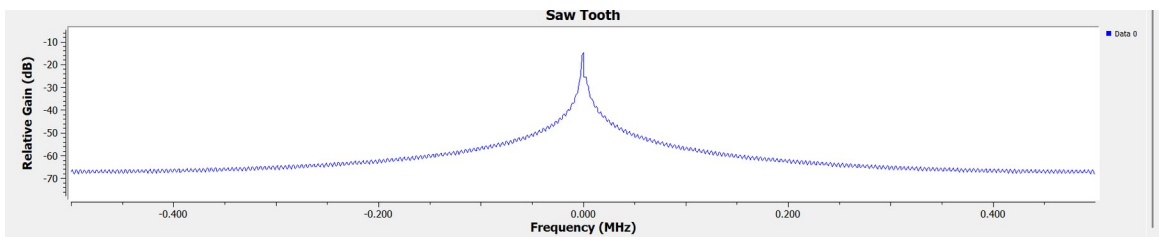


Fig. 9: Transmitted spectrum of the sawtooth wave (zoomed), showing a dense series of closely-spaced harmonics decaying slowly on either side of the main tone — the richest harmonic content among the five waveforms tested, as expected from its $1/n$ harmonic envelope.

6.5 Square Wave

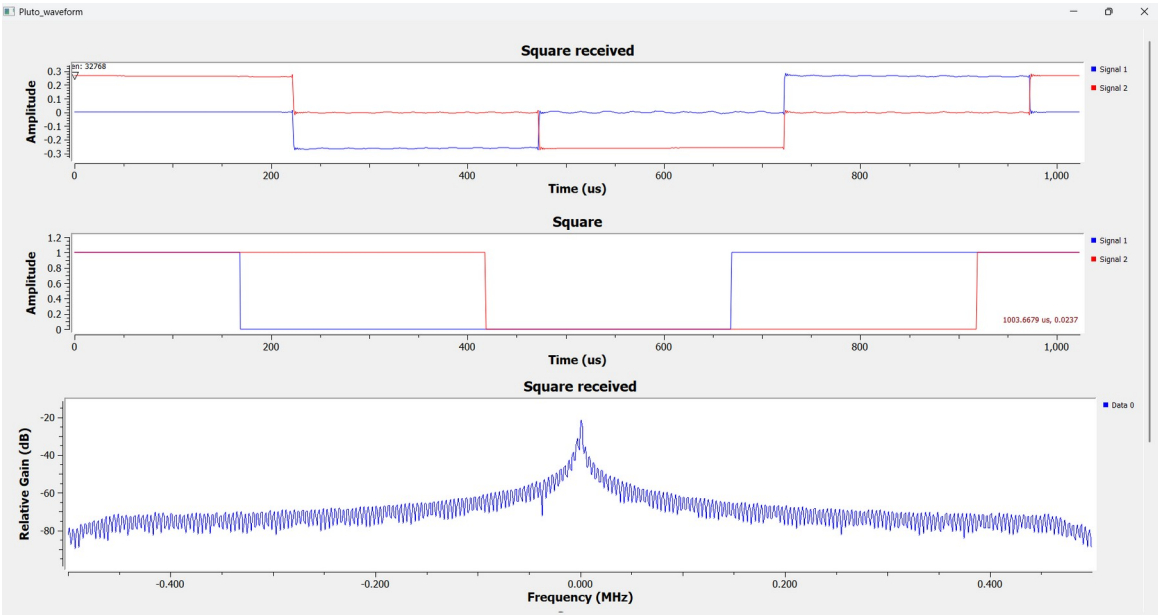


Fig. 10: Square wave — (top) received waveform showing sharp transitions between two levels, (middle) transmitted waveform, (bottom) received spectrum with a marker at 1003.6679 μ s, 0.0237 on the transmitted time-domain trace.

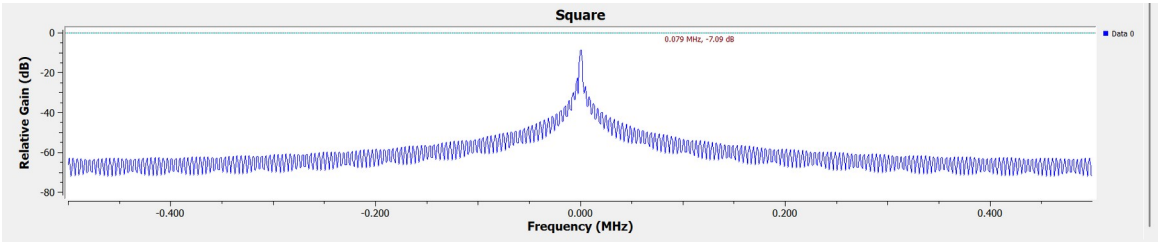


Fig. 11: Transmitted spectrum of the square wave (zoomed). Marker at 0.079 MHz, -7.09 dB. The comb of odd harmonics decays the slowest of all five waveforms, consistent with the sharp edges of a square wave requiring the richest high-frequency content after the sawtooth.

Waveform	Time-domain Shape	Spectral / Harmonic Behaviour
Cosine	Smooth single-tone oscillation	Single dominant tone; near-flat noise floor elsewhere
Sine	Smooth single-tone oscillation (90° shifted)	Single dominant tone; very deep spectral null near 0.25 MHz
Triangular	Linear rise and fall (symmetric ramp)	Main tone with fast-decaying odd-harmonic side-lobes ($\propto 1/n^2$)
Sawtooth	Linear ramp with sharp fly-back	Dense comb of many harmonics, slowly decaying ($\propto 1/n$)
Square	Sharp transitions between two fixed levels	Comb of odd harmonics, slowest-decaying / highest harmonic energy

7. Results & Discussion

All five waveforms — Cosine, Sine, Triangular, Sawtooth and Square — were successfully generated in GNU Radio, transmitted using the PlutoSDR (LO Frequency 2.4 GHz, sample rate 1 MS/s, TX attenuation 10 dB) and received back using a second PlutoSDR configured with Slow-Attack gain and 20 MHz RF bandwidth. In every case the received time-domain waveform (top trace in each figure) preserved the characteristic shape of the transmitted waveform (middle trace), confirming that the SDR link was correctly configured and operating.

The frequency-domain observations closely matched theoretical expectation. The Cosine and Sine waves, being pure single-tone signals, produced the cleanest spectra with only one dominant peak and the lowest overall harmonic energy (Figs. 2–5). The Triangular wave showed a moderate number of odd-harmonic side-lobes decaying fairly quickly on either side of the main tone (Figs. 6–7), consistent with its $1/n^2$ harmonic envelope. The Sawtooth wave exhibited the densest and most slowly-decaying comb of harmonics of all five waveforms (Figs. 8–9), consistent with it containing every harmonic (both even and odd) with a $1/n$ roll-off. The Square wave showed a comb of only odd harmonics, decaying more slowly than the triangular wave's but sparser than the sawtooth's, and with the highest peak-to-sideband harmonic energy among all the waveforms (Figs. 10–11) due to its sharp rising/falling edges.

Overall, the harmonic richness observed followed the expected order: Sawtooth > Square > Triangular > Sine $\approx$ Cosine, verifying the classical Fourier-series properties of these standard waveforms through an actual over-the-air PlutoSDR transmit-receive experiment.

8. Conclusion

Sine, Cosine, Triangular, Sawtooth and Square waveforms were successfully generated, transmitted and received using a single GNU Radio flowgraph and a PlutoSDR-based transmit-receive link, simply by changing the Waveform parameter of the Signal Source block. The received waveforms in both the time and frequency domains matched their expected theoretical shapes and harmonic content, confirming correct operation of the SDR transmit/receive chain and verifying the classical relationship between a periodic waveform's shape and its harmonic spectrum.